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Stat 50

Hw #17

Section 5.2

① $t_{n-1, \alpha/2}$

② 90% $n = 12$

$t_{11}, 0.05 = 1.796$

③ 95%, $n = 7$

$t_6, 0.025 = 2.447$

④ 99%, $n = 2$

$t_1, 0.005 = 63.657$

⑤ 95%, $n = 29$

$t_{28}, 0.025 = 2.048$

⑥

$\bar{x} = \frac{1846.14}{9} = 205.13$

$\bar{x} \pm t \times \frac{s}{\sqrt{n}}$

$205.13 \pm 2.306 \rightarrow (\frac{\sigma}{\sqrt{n}})$

$\neq \sigma = 0.96205$

$205.13 \pm 2.306 \left(\frac{0.96205}{\sqrt{9}} \right)$

$(204.39, 205.87)$

yes, there are no outliers. A 95% confidence interval is (204.32, 205.87)

(8) $\bar{x}$ = 196.64 , S = 0.68 k

(a) 95% CI

$t_{5, 0.025} = 2.571$

$196.64 \pm 2.571 \left(\frac{.68}{\sqrt{6}}\right)$

(195.93 , 197.35)

We are 95% confident that true mean temperature is between 195.93 k and 197.35

(b) 98% CI

$t_{5\ 0.01} = 3.365$

$196.64 \pm 3.365 \left(\frac{.68}{\sqrt{6}}\right)$

(195.71 , 197.57)

weare 98% confident that true mean temperature is between 195.71 k and 197.57 k.

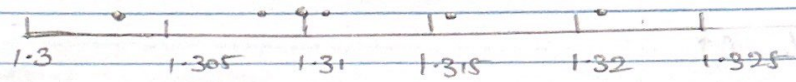
(c) $\bar{x} = 196.64$ n = 6

$\sigma = 0.00096$

Yes, the confidence intervals above be valid because they are on the sample intervals.

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(a)



(b)

$\bar{x} = 1.3115$

99%. $\sigma = 0.00626$

$n = 6$

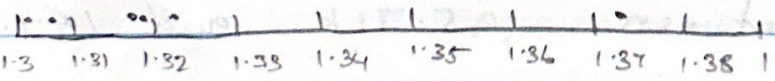
$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$1.3115 \pm 4.032 \left(\frac{\sigma}{\sqrt{6}}\right)$

$(1.3012, 1.3218)$

yes, the 92% confidence interval is $(1.3012, 1.3218)$

(c)



(d) No, the data set contains an outliers.

(10) $\bar{x} = 13\,\mu m$

$s = 2\,\mu m$ 99 %.

$n = 15$

$\bar{x} \pm t_{\alpha/2}\, s/\sqrt{n}$

$t_{14,\,0.005} = 2.977$

$13 \pm 2.977\left(\frac{2}{\sqrt{15}}\right)$

$(11.463,\ 14.537)$

We are 99 % confidence that the true mean is in (11.463, 14.537) interval.

(12) $\bar{x} = 2.64$ 95 %.

$s = 1.02$ $n = 15$

$\bar{x} \pm t\, s/\sqrt{n}$

$t_{14,\,0.025} = 2.145$

$2.64 \pm (2.145 * \frac{1.02}{\sqrt{15}})$

$(2.0751,\ 3.2049)$

We are 95 % confidence that the true mean is in (2.0751, 3.2049) interval.

Section 5.4

(1)

150°c
$n_1 = 80$
$\bar{x}_1 = 620$ psi
$\sigma_1 = 20$ psi

130°c
$n_2 = 95$
$\bar{x}_2 = 750$ psi
$\sigma_2 = 30$ psi

95%

$$750 - 620 \pm 1.96 \sqrt{\frac{20^2}{80} + \frac{30^2}{95}}$$

$$(122.54, 137.46)$$

(3)

$\bar{x}_1 = 313$ ·/.
$n_1 = 210$
$s_1 = 25$

$\bar{x}_2 = 233$ ·/.
$n_2 = 210$
$s_2 = 19$

$$x_1 - x_2 \pm Z^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

99% C.I.

$$313 - 233 \pm 2.576 \left(\sqrt{\frac{25^2}{210} + \frac{19^2}{210}}\right)$$

$$80 \pm 2.576 \; (2.1668)$$

$$(74.39, 85.61)$$

(5) 95 % CI.

$\bar{x}_1 = 3.64$ $\quad$ $\bar{x}_2 = 3.40$

$n_1 = 40$ $\quad$ $n_2 = 43$

$s_1 = 0.23$ $\quad$ $s_2 = 0.28$

$$\bar{x}_1 - \bar{x}_2 \pm Z \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$3.64 - 3.40 \pm 1.960 \left(\sqrt{\frac{0.23^2}{40} + \frac{0.28^2}{43}} \right)$$

$$.24 \pm .11161$$

$$(0.1284, 0.3516)$$

(7) $\bar{x}_1 = 105$ $\quad$ $\bar{x}_2 = 117$

$n_1 = 49$ $\quad$ $n_2 = 35$

$s_1 = 20$ $\quad$ $s_2 = 15$

98 %

$$117 - 105 \pm 2.374 \left(\sqrt{\frac{(20)^2}{49} + \frac{(15)^2}{35}} \right)$$

$$(2.934, 21.066)$$

(9) $\bar{x}_1 = 242$ $\quad$ $\bar{x}_2 = 220$

$s_1 = 20$ $\quad$ $s_2 = 31$ $\quad$ 95 %

$n_1 = 47$ $\quad$ $n_2 = 42$

$$242 - 220 \pm 2.000 \left(\sqrt{\frac{(20)^2}{47} + \frac{(31)^2}{42}} \right)$$

(1)

$= (10.820, 33.180)$

(ii) $\bar{x}_1 = 91.1$ $\quad \bar{x}_2 = 90.7$

$s_1 = 6.23$ $\quad s_2 = 4.34$ $\quad 95\%$

$n_1 = 50$ $\quad n_2 = 40$

(a) $91.1 - 90.7 \pm 1.987 \left(\sqrt{\frac{(6.23)^2}{50} + \frac{(4.34)^2}{40}} \right)$

$(-1.820, 2.620)$

(b) No, Since 0 is in the confidence interval it may be regarded as being a plausible value for the mean difference in hardness.